

HACKATHON

*AI Innovation for Resilient
Agri-Food Systems*

FAO / ITU / IFAD / WFP

Second Global Initiative Meeting on

AI for Agri-Food Systems

10 October 2026

APPLICANTS TOOLKIT

HACKATHON • 8-9 OCTOBER 2026

NASC COMPLEX, ICAR, NEW DELHI, INDIA

Diverse Local Solutions.
Scalable Impact.

Join the International Hackathon 2026

This Hackathon 2026 is an international innovation event.

This Hackathon is being organized under the umbrella of the Global Initiative on AI for Agri-Food Systems (GI-AI4AFS), which is lead by the International Telecommunication Union (ITU), United Nations World Food Programme (WFP), Food and Agriculture Organization of the United Nations (FAO) and International Fund for Agricultural Development (IFAD).

It is being co-organized in technical partnership with Department of Telecommunications (DoT) (India), the Indian Council for Agricultural Research (ICAR), Fraunhofer HHI, Balarama AgriTech , AgHub and the Accelerating Climate Resilient Agriculture in Telangana, India through Data Driven Agro Ecological TestHubs (ACRAT) Project, with support from the Federal Ministry of Agriculture, Food and Regional Identity (Germany).

We are delighted to invite students, researchers, professionals, entrepreneurs, technology experts, practitioners, and innovators from around the world to participate. The hackathon provides a collaborative platform to exchange ideas, combine diverse expertise, and co-create innovative solutions to real-world challenges in agriculture and food systems.

You do not need to be a programmer to participate. We welcome participants from different disciplines, backgrounds, and levels of experience. Bring your knowledge, creativity, and enthusiasm, connect with an international community, and work together to transform promising ideas into practical and impactful solutions.

What is a Hackathon?

A Hackathon is a time-limited collaborative innovation event where multidisciplinary teams work intensively to develop practical solutions to real-world challenges.

Participants bring together different skills and perspectives to explore a challenge, develop and test ideas, build prototypes or concepts, and present their solutions within a structured timeframe.

The International Hackathon 2026 combines online preparation and collaboration with an intensive in-person Hackathon in New Delhi, India. Selected participants will have access to challenge information, technical resources, expert guidance, mentoring, and the GI-AI4AFS Open Reference Architecture.

The Hackathon provides an opportunity to learn, collaborate, experiment, and transform innovative ideas into practical solutions for real-world challenges in AI-enabled agri-food systems.

Who Can Join the Hackathon?

The International Hackathon 2026 welcomes participants from around the world with diverse backgrounds, skills, and levels of experience. We invite students, researchers, farmers, agricultural and food-system experts, software developers, AI and data specialists, engineers, entrepreneurs, start-ups, industry professionals, policymakers, designers, and innovators interested in developing solutions for real-world challenges across agriculture and global agri-food value chains.

We particularly encourage ideas that consider the needs and perspectives of small and marginal farmers, including solutions that are accessible, affordable, practical, and relevant to local farming conditions. Participants are encouraged to explore how AI and digital technologies can support farmers while strengthening connections across the food value chain, from production and resource management to aggregation, processing, logistics, markets, and consumers.

You do not need to be a programmer to participate. The Hackathon encourages multidisciplinary collaboration that combines technical expertise, agricultural knowledge, farmer perspectives, business and value-chain expertise, sustainability, policy, and user-centered design.

Team Composition

Participants are encouraged to form multidisciplinary teams with complementary skills and expertise. Ideally, each team should bring together a mix of:

- Technology expertise, such as AI, data science, software development, IoT, digital platforms, or data engineering.
- Agriculture and food-system expertise, including agronomy, farming, agricultural engineering, natural-resource management, or food systems.
- Farmer and field perspectives, particularly an understanding of the needs and realities of small and marginal farmers.
- Business and value-chain expertise, including entrepreneurship, markets, logistics, processing, finance, and global agri-food value chains.
- Design, sustainability, and policy perspectives to support usability, inclusiveness, responsible innovation, and scalability.

Not every team member needs to have technical or programming skills. Strong teams combine technology with domain knowledge and real-world perspectives, ensuring that proposed solutions address genuine needs and can be practically implemented.

Teams are encouraged to develop inclusive, interoperable, and scalable solutions that create value for small and marginal farmers while contributing to more sustainable, resilient, and connected global agri-food value chains.

Hackathon Challenges

The International Hackathon 2026 invites teams to address six interconnected challenges facing global agri-food systems. Across all challenges, teams are encouraged to avoid isolated data silos and standalone platforms. Solutions should build on the GI-AI4AFS Open Reference Architecture, use open interfaces where possible, and demonstrate how locally relevant innovations can become interoperable and scalable building blocks.

The goal is not to create another app for farmers. The goal is to make data, AI, knowledge, and digital services work together, while keeping the experience simple and useful for farming community and value chain actors.

1. Smart and Climate-Resilient Agriculture

Develop AI-driven solutions that improve farm productivity, strengthen climate resilience, and promote sustainable use of natural resources.

Examples: Precision agriculture; smart irrigation and water management; soil health monitoring; climate adaptation; carbon-smart farming; resource optimization; smart farm management.

2. Crop Intelligence and Decision Support

Leverage AI and agricultural data to enable timely, accurate, and informed decisions throughout the crop production cycle.

Examples: Pest and disease detection; crop health diagnostics; drone and satellite-based crop monitoring; yield forecasting; weather and early-warning systems; AI-powered decision-support tools.

3. Post-Harvest Management and Value Addition

Develop innovative solutions that minimize post-harvest losses, maintain food quality and safety, and enhance value addition.

Examples: Smart storage and warehousing; quality and safety assessment; intelligent grading and sorting; processing automation; shelf-life optimization; value-added food processing.

4. Digital Agricultural Supply Chains

Strengthen the efficiency, transparency, resilience, and competitiveness of agricultural value chains through AI-enabled digital solutions.

Examples: Supply-chain optimization; cold-chain monitoring; logistics and distribution; digital marketplaces; market intelligence; food traceability; export readiness; supply-chain risk management.

5. Farmer-Centric Digital Services

Develop inclusive AI-enabled services that improve farmers' access to knowledge, finance, markets, and risk-management mechanisms, particularly for small and marginal farmers.

Examples: Personalized agricultural advisory; agricultural credit scoring; crop insurance; financial services; market access; multilingual and accessible digital tools; inclusive solutions for women and youth.

6. Sustainable and Circular Agri-Food Systems

Design AI-enabled solutions that accelerate the transition toward sustainable, resource-efficient, low-waste, and circular agri-food systems.

Examples: Circular bioeconomy; agricultural waste-to-wealth; renewable energy applications; by-product utilization; biodiversity conservation; resource recovery and recycling.

ONE GLOBAL ARCHITECTURE. Diverse Local Solutions. Scalable Impact.

Teams are encouraged to build on the GI-AI4AFS Open Reference Architecture, using available APIs, open interfaces, reference implementations, and GitHub resources to create interoperable and reusable solutions rather than isolated platforms or silos.

As a cross-cutting priority, all solutions should follow principles of responsible, trusted and human-centred AI, including transparency, human oversight, privacy, data governance, fairness, security, accountability, and interoperability. Attention should be given to solutions that are accessible, trustworthy, and beneficial to small and marginal farmers, while respecting their rights, data, knowledge, and local contexts.

Application Process

Interested participants and teams should complete the official [<https://forms.gle/w6GAzdxnwYouwLhr5>] within the announced registration period.

Applicants are not expected to have a complete product or fully developed prototype at this stage.

What Should You Submit?

Applicants or teams should provide:

- Applicant and team information, including affiliation, country, background, and areas of expertise.
- Selected Hackathon challenge.
- Problem statement describing the real-world agri-food challenge.
- Proposed solution or initial concept.
- Target users or beneficiaries, including small and marginal farmers where relevant.
- Team expertise and capabilities.
- Initial technical approach, including the proposed use of AI, digital technologies, IoT, or other relevant technologies.
- Expected impact for farmers, food systems, or agri-food value chains.

- Interoperability potential, including how the solution could connect with existing services, APIs, or the GI-AI4AFS Open Reference Architecture.
- Motivation to participate in the International Hackathon 2026.

A functional prototype or final pitch deck is not required at the application stage.

Selection Process

Applications will be evaluated by an international panel of experts based on:

- Innovation and originality
- Relevance to agricultural value chains
- Technical feasibility
- Potential impact
- Scalability
- Sustainability
- Inclusiveness
- Quality of the proposed solution

Selected applicants from around the world will be invited to participate in an intensive two-day International Hackathon Boot Camp in New Delhi, India.

What Should Your Team Deliver?

At the end of the Hackathon, each team should submit:

- A pitch deck presenting the problem, proposed solution, technical approach, feasibility, and potential impact.
- A prototype, wireframe, source code, or functional demonstration showing how the solution works.
- Technical documentation and code, where applicable, shared through the Hackathon GitHub environment.

Teams are encouraged to develop solutions that are practical, demonstrable, interoperable, and reusable. Solutions that integrate with the GI-AI4AFS Open Reference Architecture and use open standards and open-source approaches will be viewed positively during evaluation.

Solutions developed during the Hackathon will remain the **intellectual property of the respective teams**, subject to the Hackathon terms and conditions.

Final Evaluation

Solutions will be evaluated by an international jury of experts based on:

- Innovation and originality
- Relevance to the challenge
- Quality and functionality of the prototype
- Technical feasibility
- Farmer and user value
- Interoperability and integration potential
- Responsible and trustworthy AI
- Inclusiveness and accessibility
- Potential impact
- Sustainability
- Scalability and reusability
- Quality of the final pitch and demonstration

Teams should demonstrate who the solution benefits, how it works, how it connects with the broader ecosystem, and how it could progress beyond the Hackathon.

Final Showcase and Awards

The hackathon culminates in a live Grand Finale, where shortlisted teams will pitch their solutions before an international jury of experts from academia, government, international organizations, industry, the investment community, and the innovation ecosystem. The winning teams will be invited to showcase their innovations during the ITU, FAO, IFAD & WFP Second Global Initiative Meeting on AI for Agri-Food Systems, to be held in New Delhi, presenting to policymakers, agricultural scientists, AI experts, development practitioners, private-sector leaders, and investors.

Awards: [Prize structure to be announced – e.g. cash prizes and/or sponsored awards per theme.]

Beyond recognition, participants can build strategic partnerships and explore pathways for piloting, scaling, commercialization, and adoption. Outstanding solutions may be considered for mentorship, incubation, research collaboration, and partnership opportunities through the organizing institutions and their networks. In particular, solutions that build on the open reference architecture may enter the Global Initiative's structured use-case selection and be considered for a funded pilot implementation on the reference architecture in 2027; solutions considered for such piloting are expected to be made available under an open licence.

Key Dates

Date	Event / Activity
15 September 2026	Registration opens
2 October 2026	Registration deadline

Platforms and Tools

Throughout the Hackathon, participants will have access to a range of digital platforms, technical resources, and collaboration tools to support teamwork and solution development.

- GitHub: Access to Hackathon repositories, technical documentation, sample code, APIs, datasets, and other development resources.
- GI-AI4AFS Open Reference Architecture: A common technical foundation for developing interoperable solutions and connecting data, AI services, and applications.
- Datasets and APIs: Selected datasets, open interfaces, and sample data will be provided to support prototype development and testing.
- Online Collaboration Platform: Used for virtual sessions, team collaboration, mentoring, and communication during the Boot Camp and Hackathon.
- Development Tools: Teams may use suitable open-source frameworks, AI tools, programming environments, and other technologies required to develop their solutions.

Detailed access instructions and technical guidance will be provided to selected participants before the Boot Camp and Hackathon

Practical Requirements

Participation in the International Hackathon 2026 is free of charge. [To be confirmed]

Participants should have:

- A laptop or computer suitable for solution development.

- A stable internet connection for online activities.
- A GitHub account, where required.
- Access to designated collaboration and communication platforms.
- Availability for scheduled online preparation, Hackathon activities, mentoring, and final presentations.
- Relevant software or development tools, where required.
- For in-person participation, the necessary travel documents and arrangements to attend the event in New Delhi, India.

Code of Conduct

The Hackathon 2026 is committed to a respectful, inclusive, collaborative, and safe environment.

Participants are expected to:

- Treat others with respect, professionalism, and fairness.
- Support diversity, inclusion, collaboration, and open exchange of ideas.
- Respect the knowledge, rights, perspectives, and data of farmers and local communities.
- Use AI, information, software, and other resources responsibly, legally, and ethically.
- Respect intellectual property, licensing, privacy, confidentiality, and data protection.
- Properly acknowledge external software, AI models, tools, and other resources.
- Refrain from harassment, discrimination, plagiarism, misuse of information, or harmful behaviour.
- Follow the instructions and decisions of the Hackathon organisers.

Serious violations may result in disqualification or removal from the Hackathon, subject to the organisers' rules and procedures.

By participating, all participants agree to contribute to a responsible, inclusive, respectful, and collaborative innovation environment.

Contact and Enquiries

For questions regarding applications, eligibility, participation, travel, or practical arrangements:

General Enquiries:

raghu.chaliganti@hhi.fraunhofer.de

rabi.sahoo@icar.org.in

Secretariat of the Global Initiative on AI for Agri-Food Systems

ai4food@itu.int

For questions regarding the GI-AI4AFS Open Reference Architecture, GitHub resources, APIs, datasets, or the development environment:

Technical Enquiries:

jan.redlich@bonnconsulting.group